

Combinational Logic

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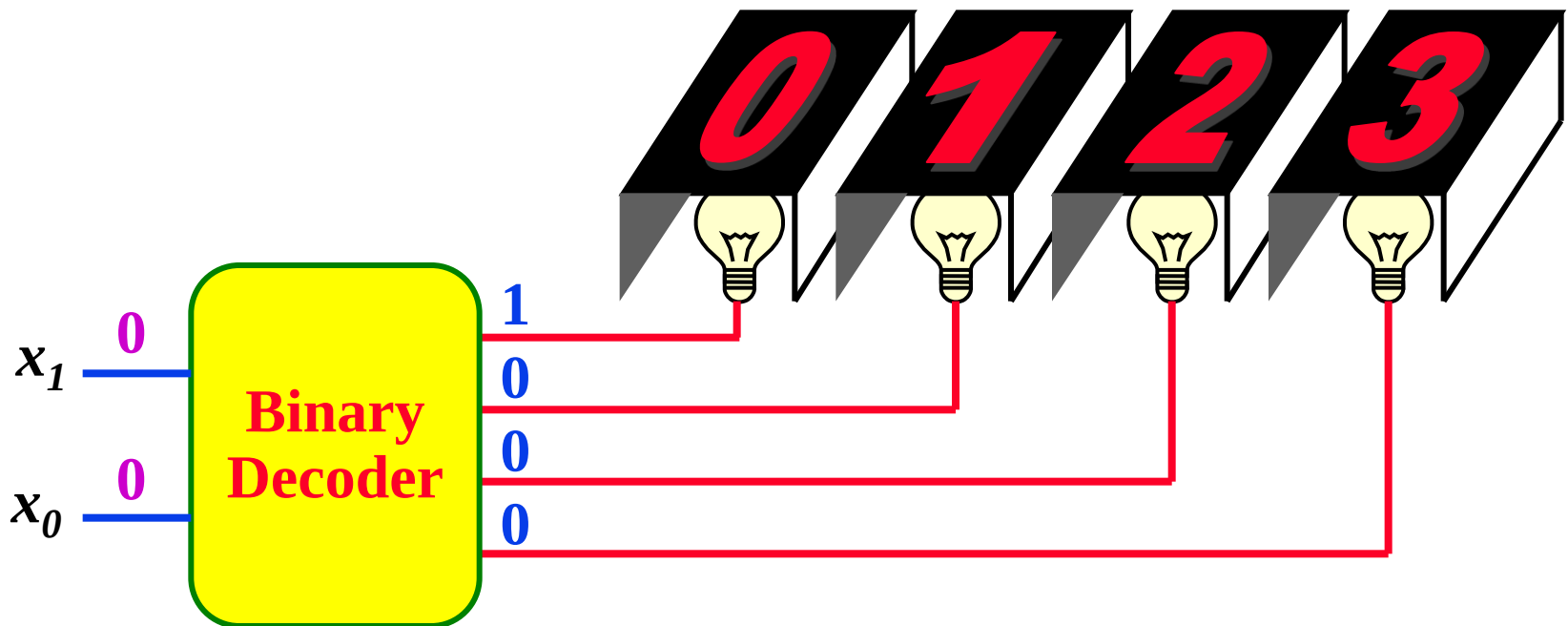
Decoders

★ Extract “*Information*” from the code

★ Binary Decoder

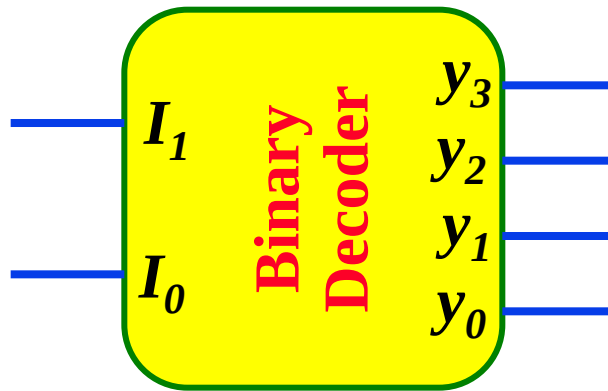
- Example: 2-bit Binary Number

Only *one* lamp will turn on

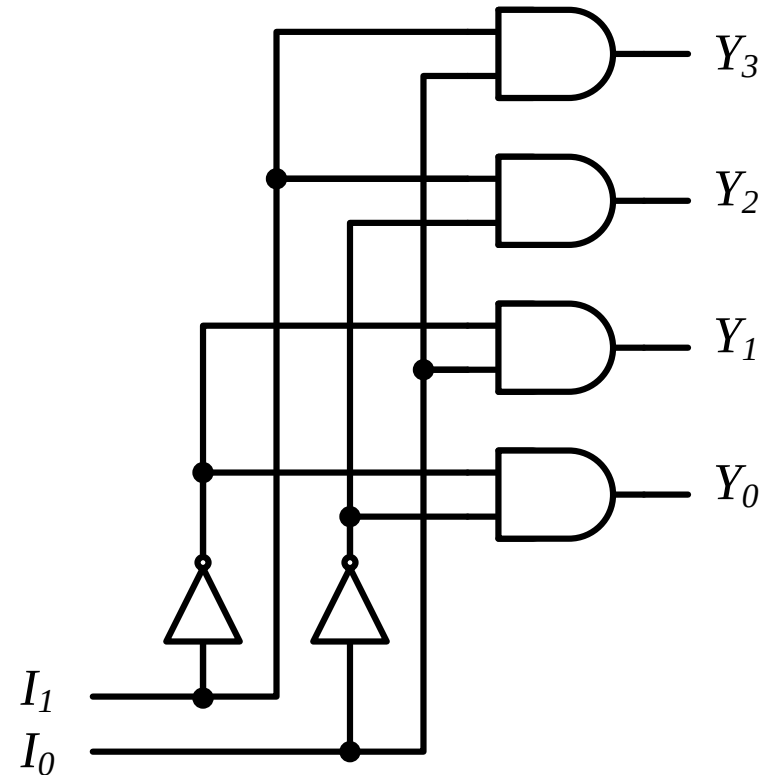


Decoders

★ 2-to-4 Line Decoder



I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



$$Y_3 = I_1 I_0$$

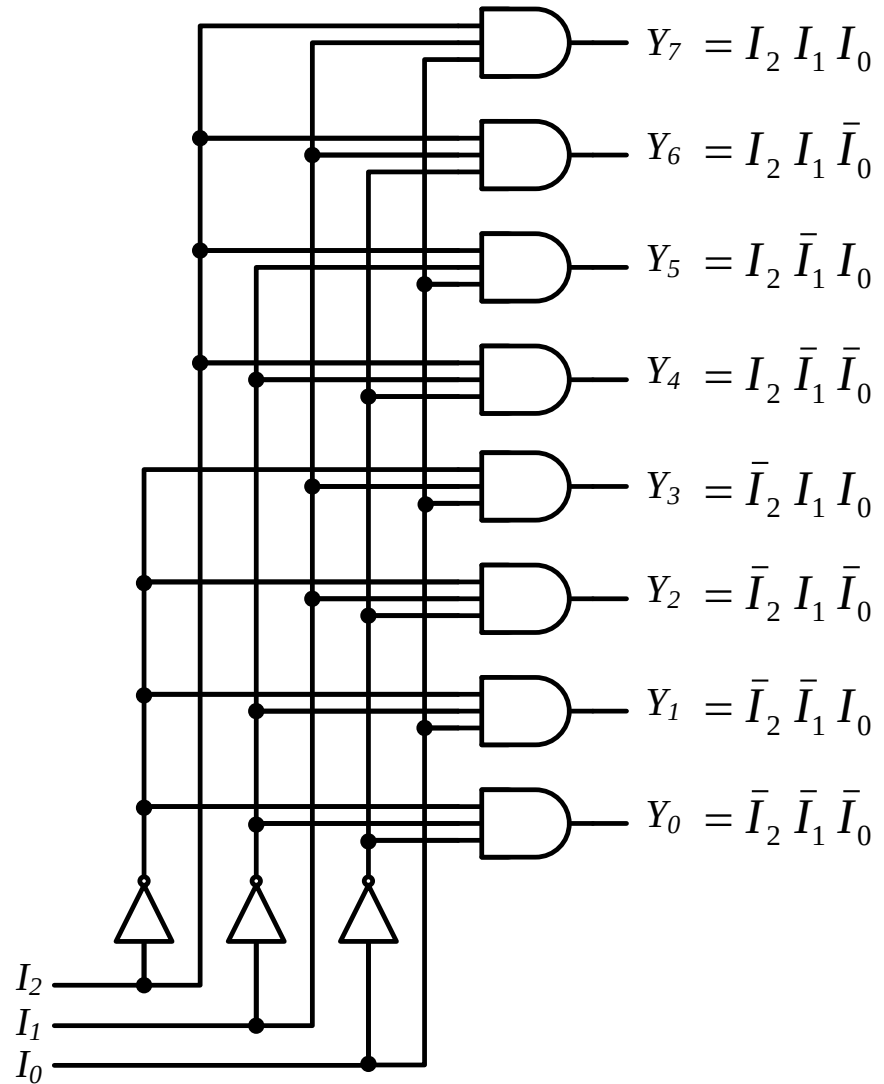
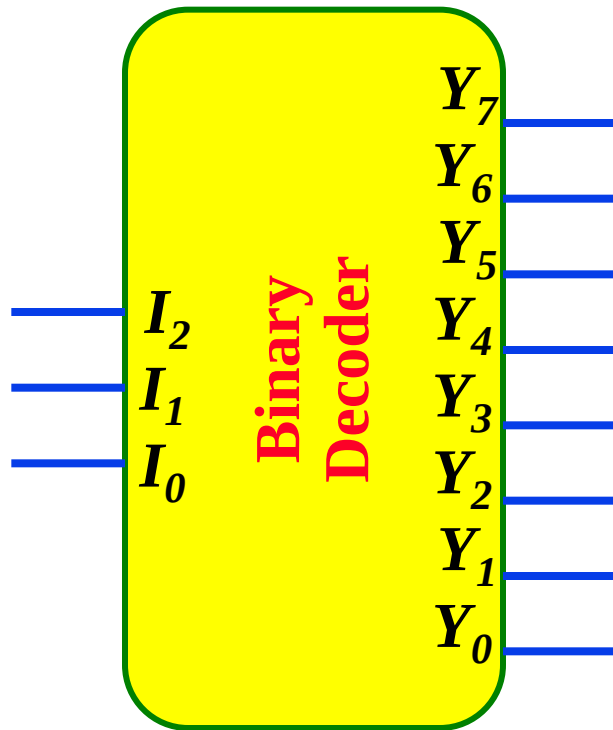
$$Y_1 = \bar{I}_1 I_0$$

$$Y_2 = I_1 \bar{I}_0$$

$$Y_0 = \bar{I}_1 \bar{I}_0$$

Decoders

★ 3-to-8 Line Decoder

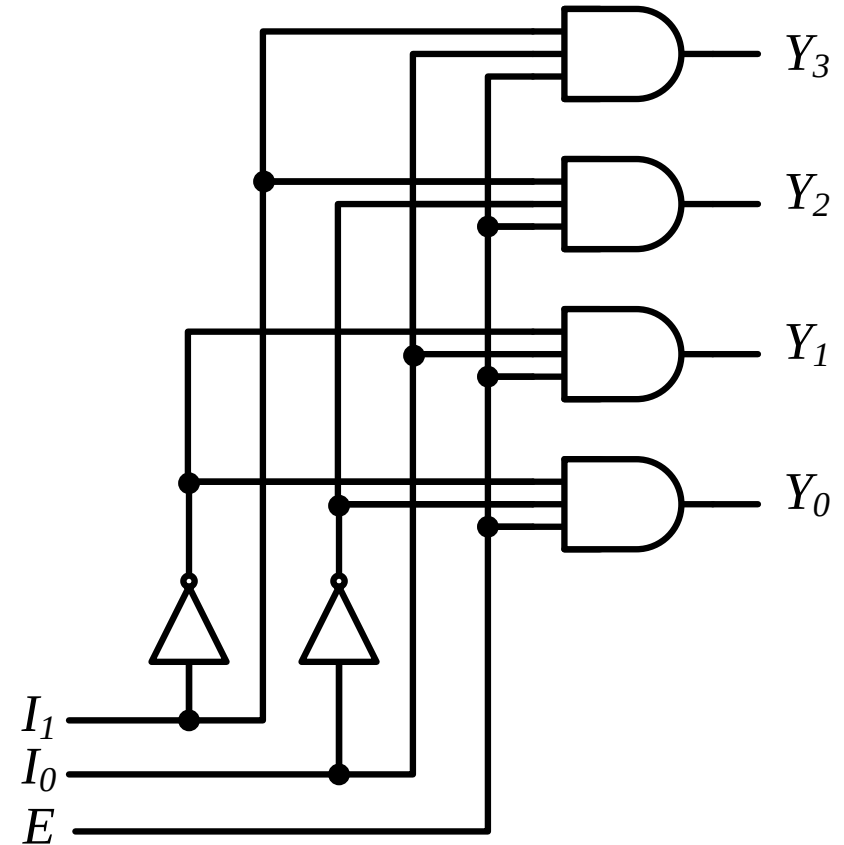


Decoders

★ “Enable” Control



E	I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

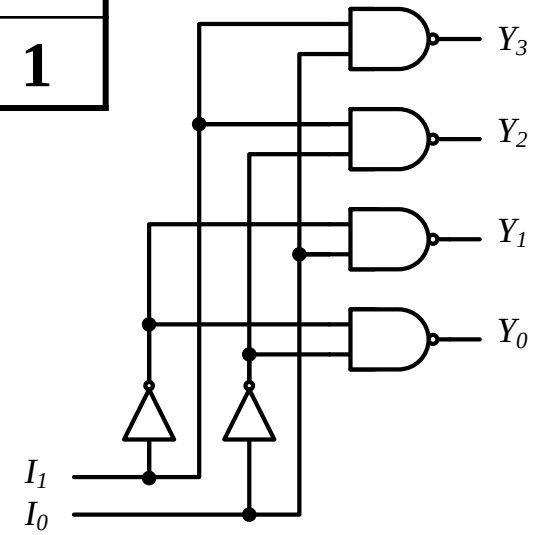
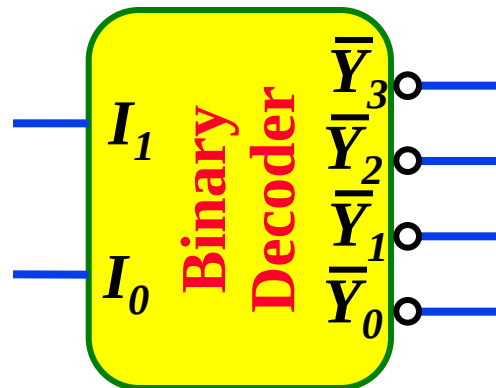
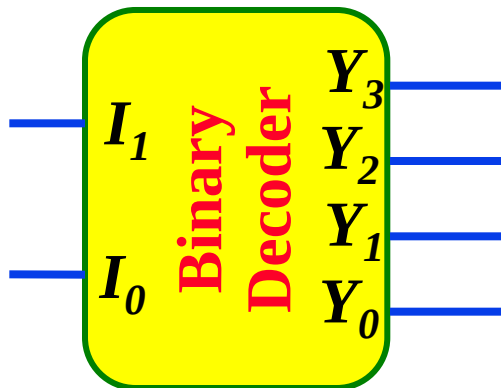


Decoders

★ Active-High / Active-Low

I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	1



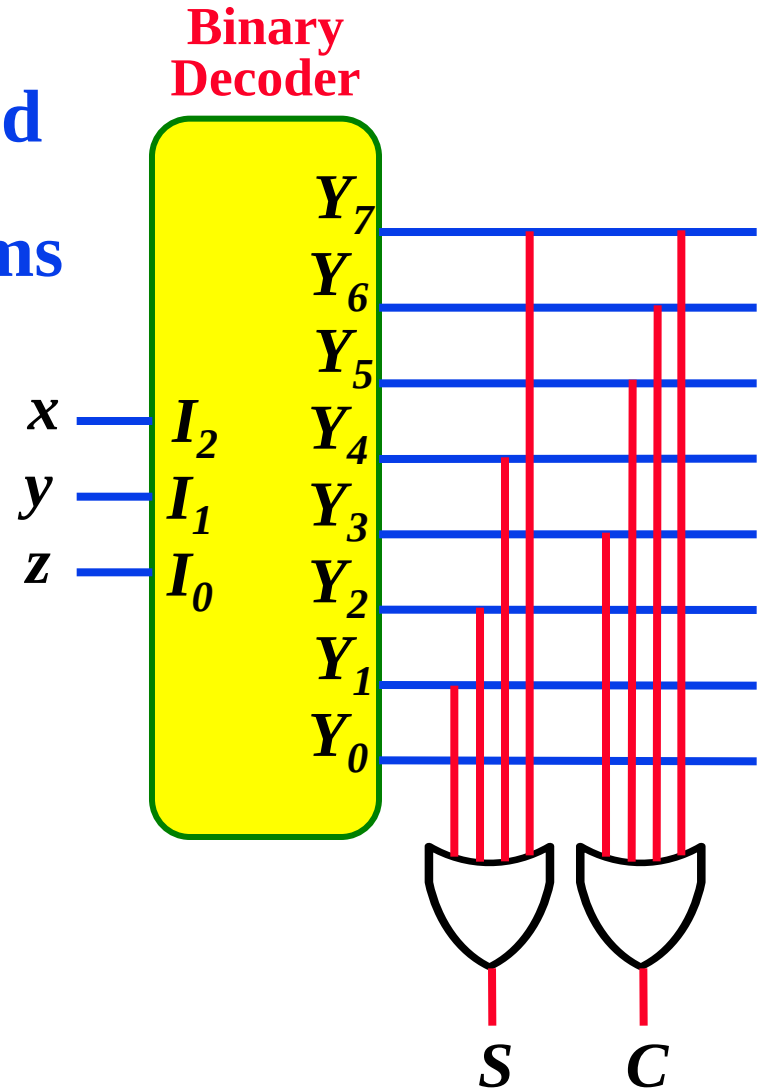
Implementation Using Decoders

- ★ Each output is a minterm
- ★ All minterms are produced
- ★ Sum the required minterms

Example: Full Adder

$$S(x, y, z) = \sum(1, 2, 4, 7)$$

$$C(x, y, z) = \sum(3, 5, 6, 7)$$

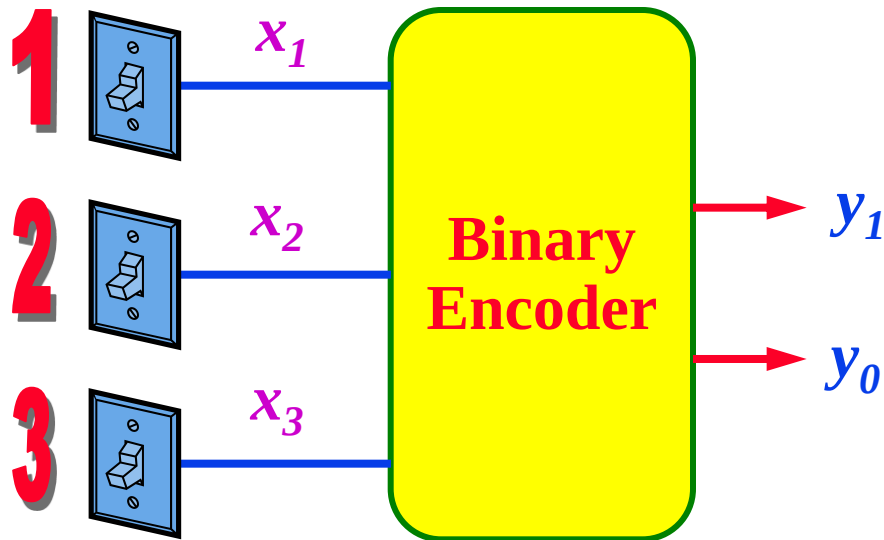


Encoders

★ Put “*Information*” into code

★ Binary Encoder

- Example: 4-to-2 Binary Encoder



Only *one* switch should be activated at a time

x_3	x_2	x_1	y_1	y_0
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
1	0	0	1	1

Encoders

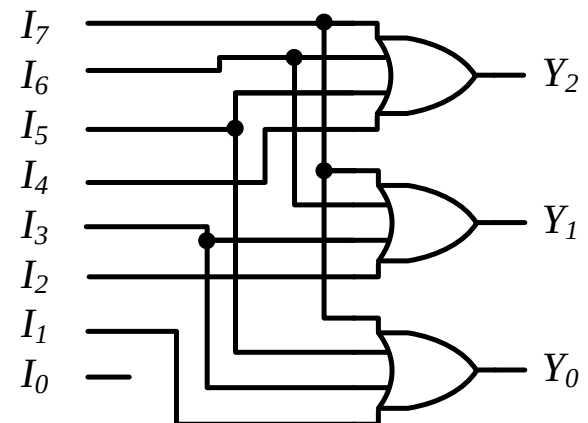
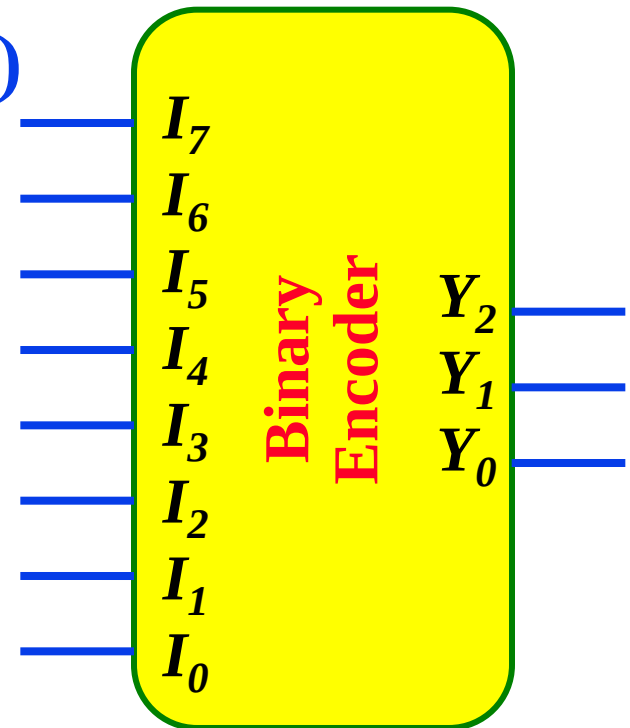
★ Octal-to-Binary Encoder (8-to-3)

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

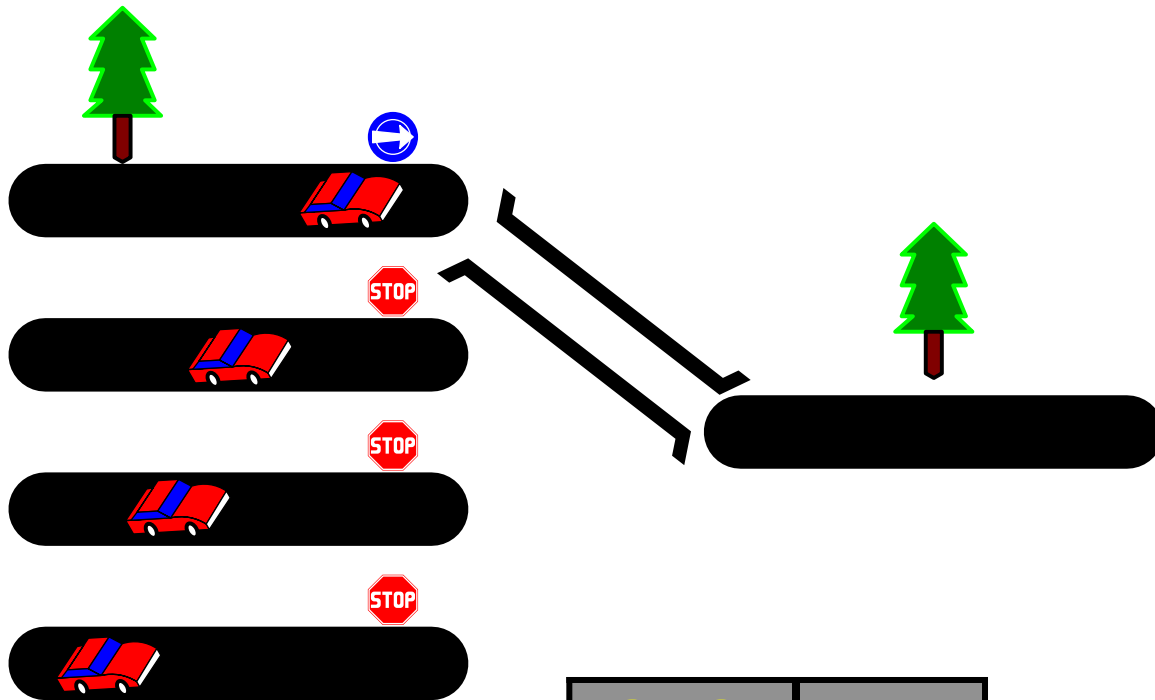
$$Y_2 = I_7 + I_6 + I_5 + I_4$$

$$Y_1 = I_7 + I_6 + I_3 + I_2$$

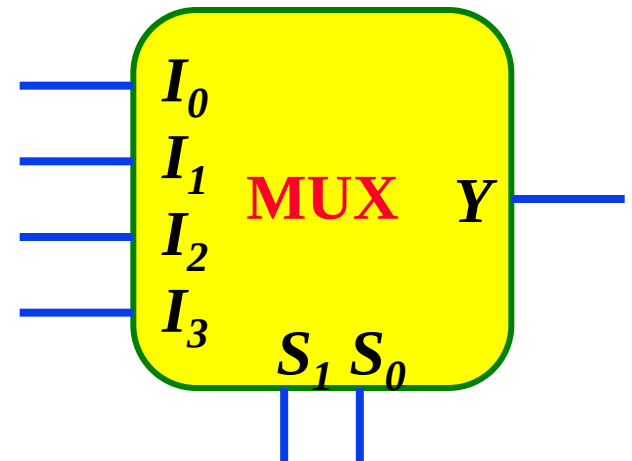
$$Y_0 = I_7 + I_5 + I_3 + I_1$$



Multiplexers

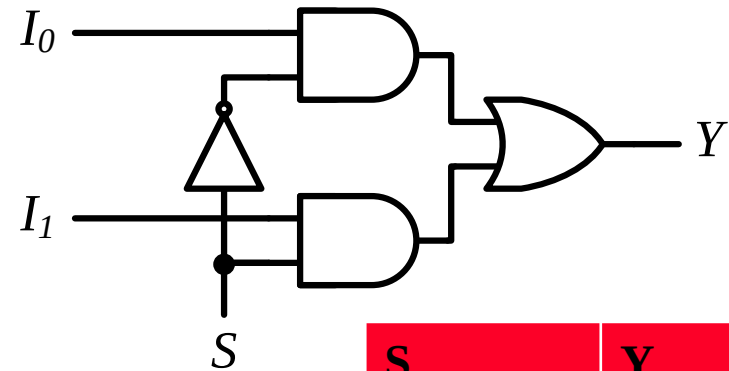
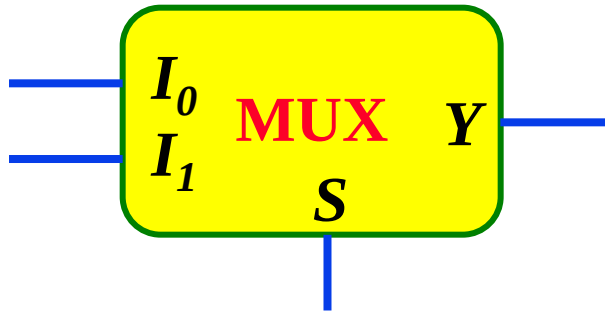


S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



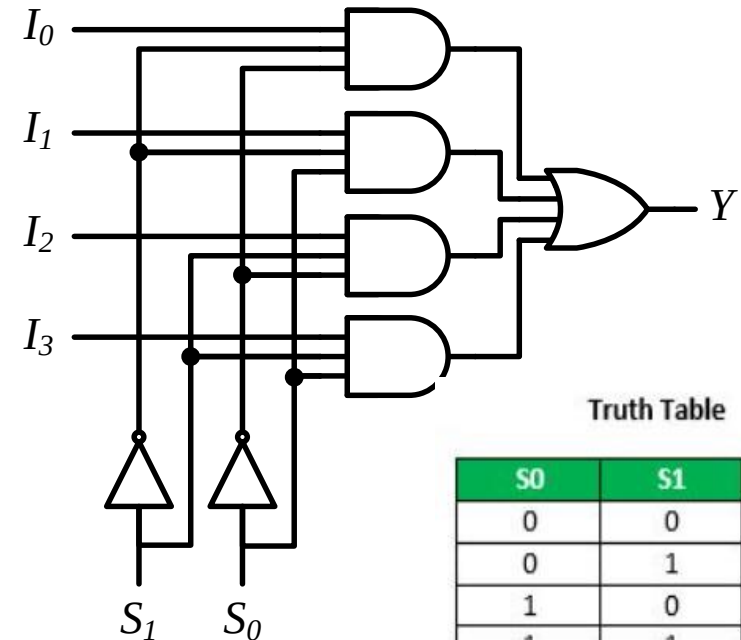
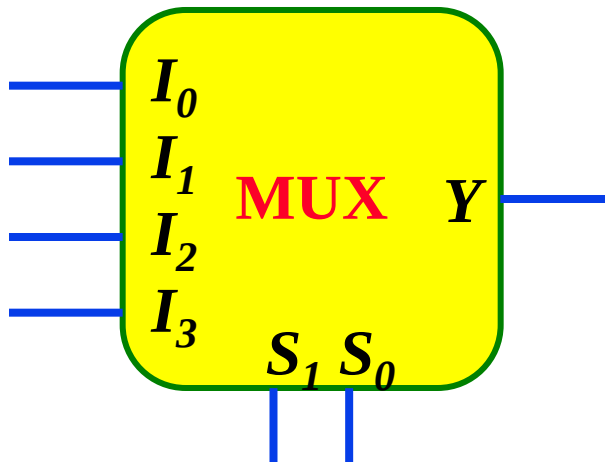
Multiplexers

★ 2-to-1 MUX



S	Y
0	I_0
1	I_1

★ 4-to-1 MUX



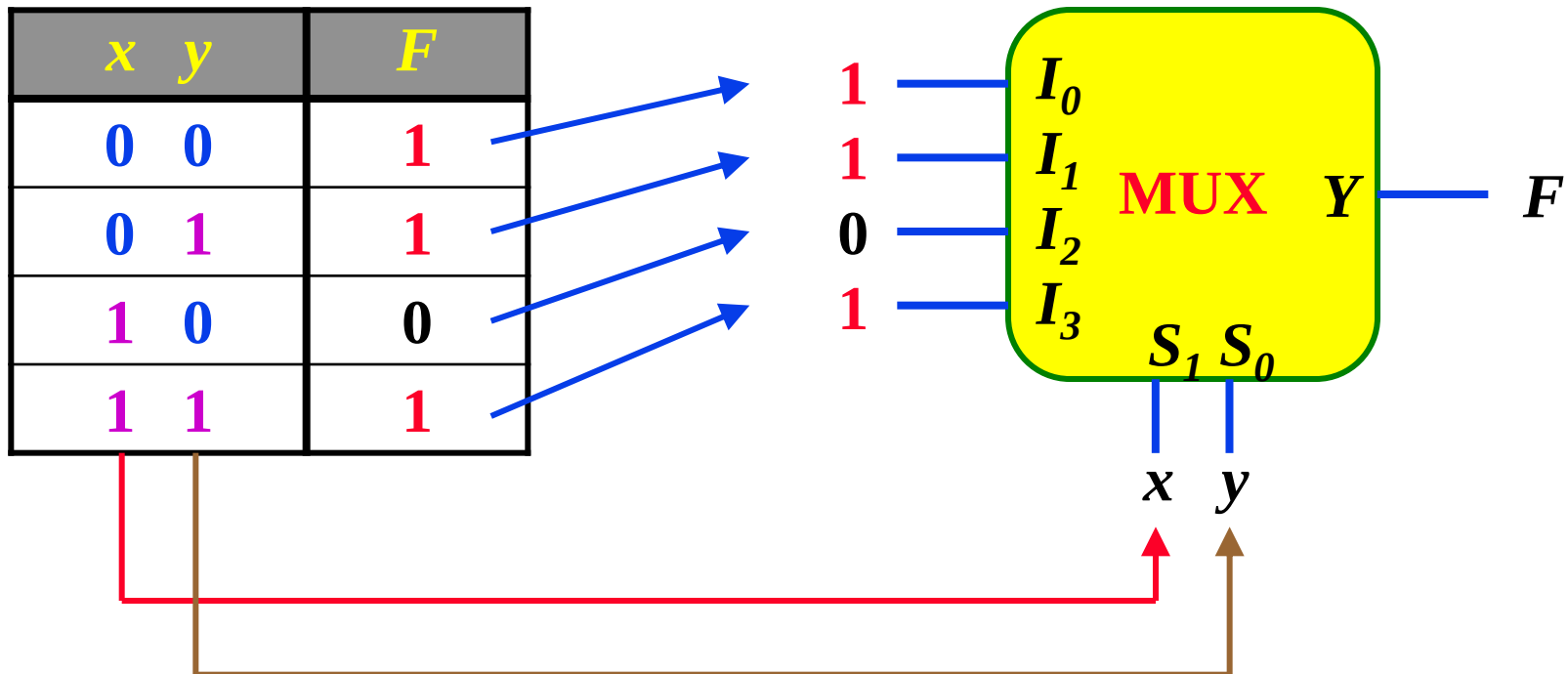
Truth Table

S_0	S_1	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Implementation Using Multiplexers

★ Example

$$F(x, y) = \sum(0, 1, 3)$$

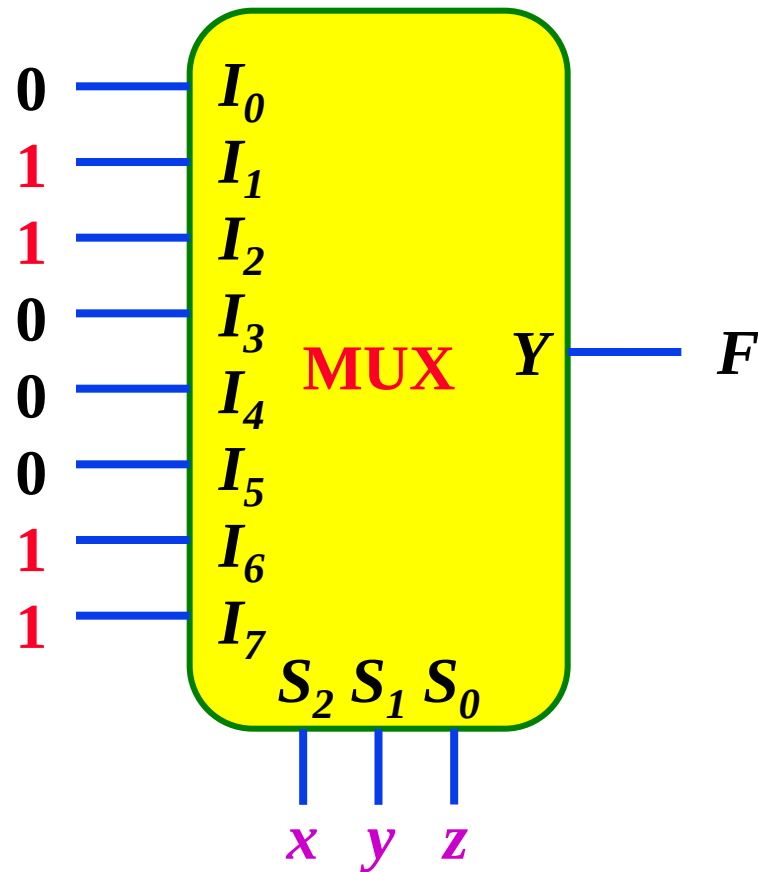


Implementation Using Multiplexers

★ Example

$$F(x, y, z) = \sum(1, 2, 6, 7)$$

<i>x</i>	<i>y</i>	<i>z</i>	<i>F</i>
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1



Implementation Using Multiplexers

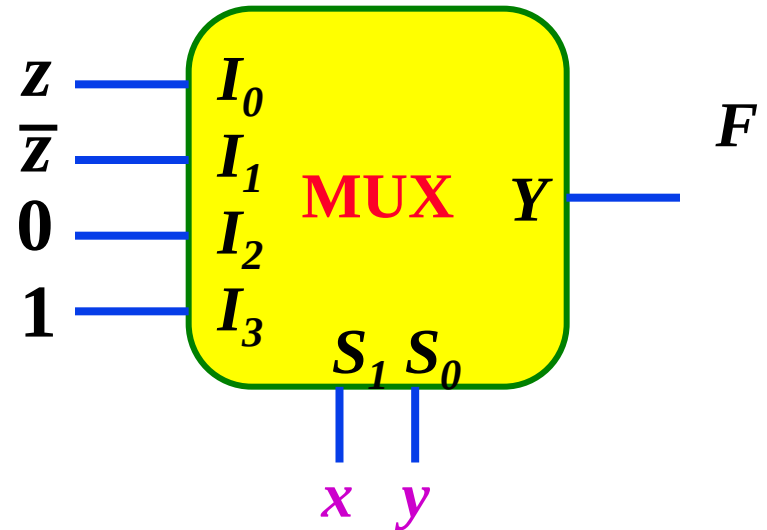
★ Example

$$F(x, y, z) = \sum(1, 2, 6, 7)$$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Implementation rules from truth table:

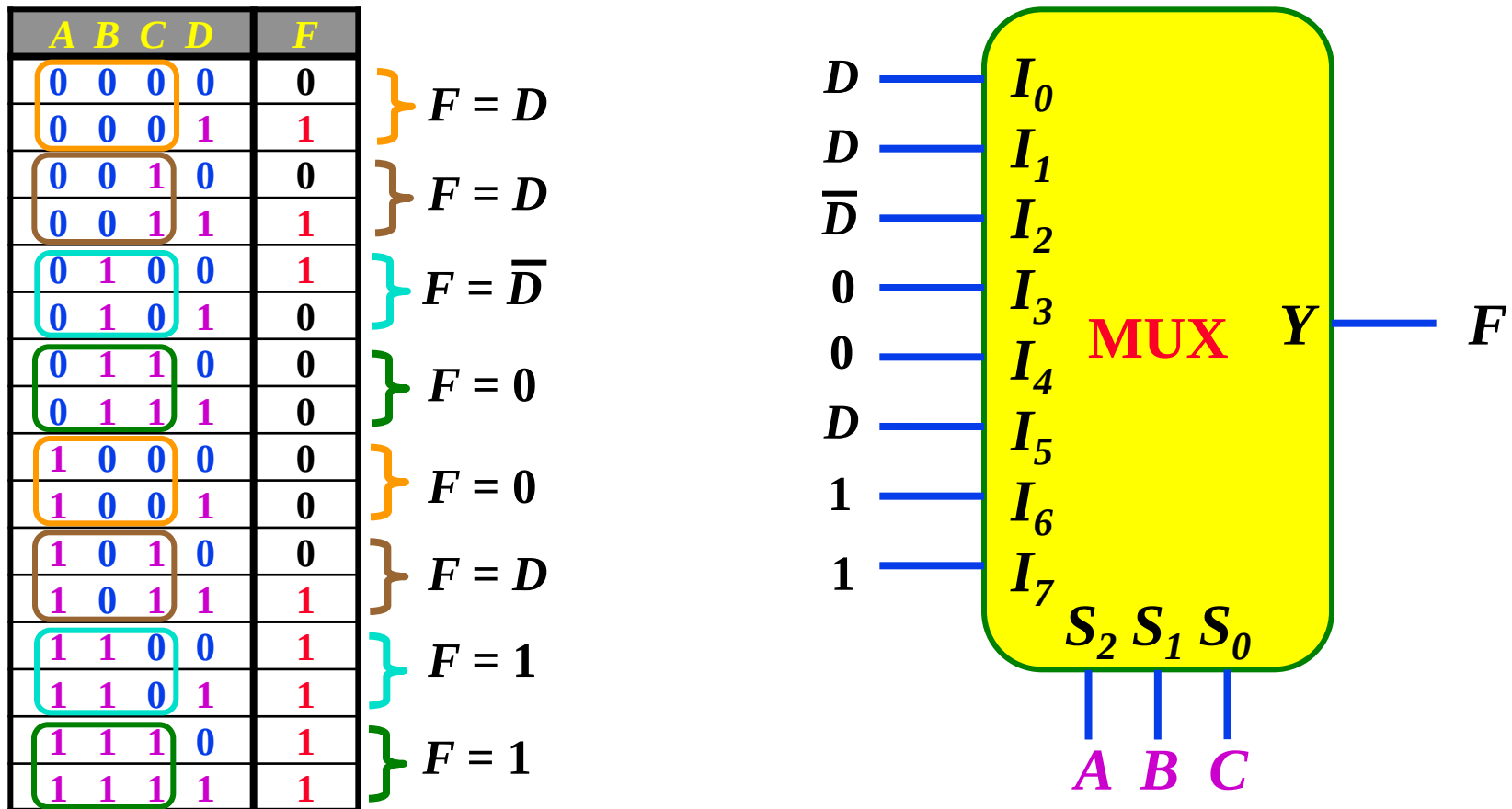
- $F = z$ (for $x=0, y=0$)
- $F = \bar{z}$ (for $x=0, y=1$)
- $F = 0$ (for $x=1, y=0$)
- $F = 1$ (for $x=1, y=1$)



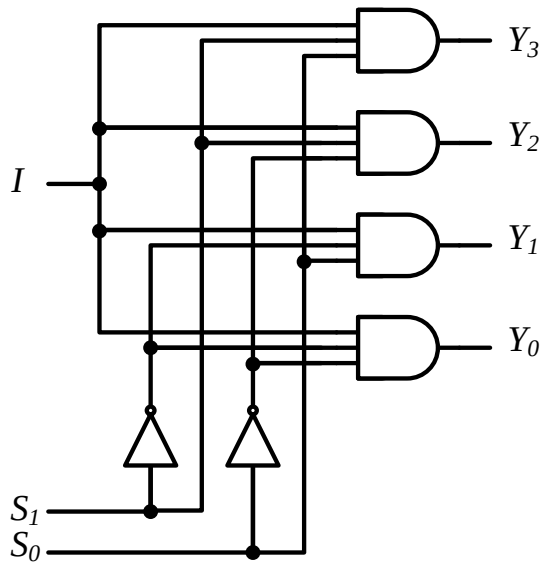
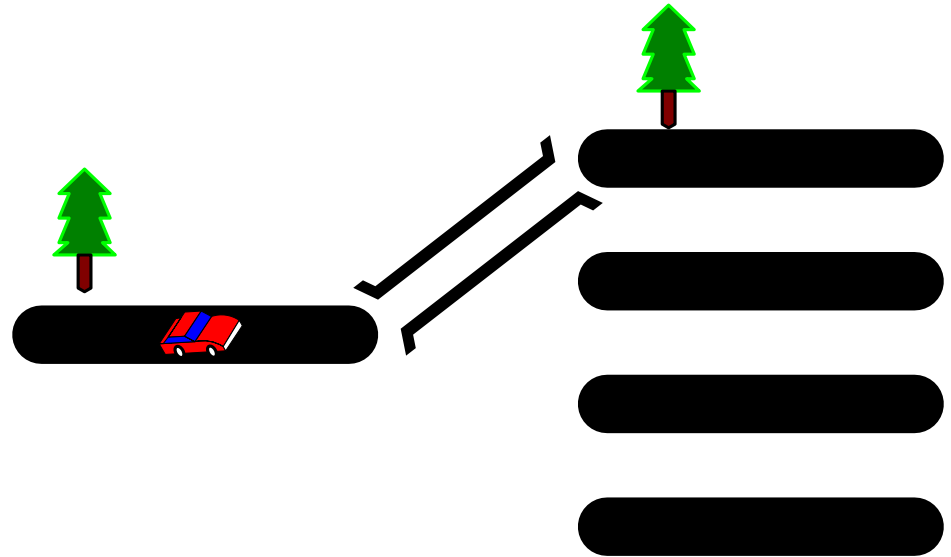
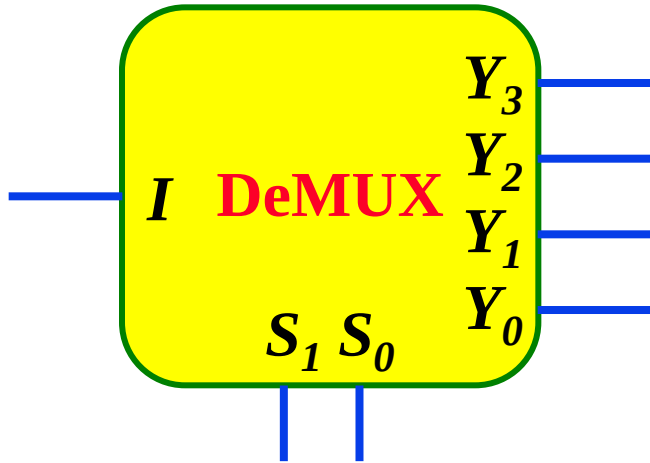
Implementation Using Multiplexers

★ Example

$$F(A, B, C, D) = \sum(1, 3, 4, 11, 12, 13, 14, 15)$$



DeMultiplexers



S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0